METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

Chulalongkorn University · Department of Chemical Engineering

PROCUREMENT PACKAGE LIST

R-201 synthesis loop · 14 packages · scope, documents, inspection and test, lead time, criticality

DOCUMENT NO. : MEOH-V1-PM-PKG-001

DOCUMENT TITLE : Procurement Package List

PROJECT : Methanol Converter Ver 1 · OTS

STATUS : Rev 0 · Issued for teaching

Teaching document generated from the Methanol Converter Ver 1 simulator model; numbers come from the model and the OTS tabs. Not for construction.

Rev.	Date	Revision status	Originator	Checker	Approver
0	27-Oct-26	Issued for teaching	A. Soottitantawat		

1 Purpose and scope

This list divides the equipment and materials of the Methanol Converter Ver 1 synthesis loop (R-201, C-201, E-201 ... E-204, V-201 and their instruments, valves, SIS, relief devices, piping and electrical supply) into procurement packages. For each package it gives the scope with the P&ID tags, the key datasheet or document the requisition is issued against, the inspection and test plan (FAT and witness points), a typical lead time and the criticality. It also checks the lead times against the required-on-site dates of an assumed construction plan.

It is a teaching document generated from the simulator model and the EPC P&IDs MEOH-PID-001 / 002. No vendor, client or contractor is named. Lead times are typical-practice ranges for this kind of equipment (assumed), not quotations. Requisition months, required-on-site (ROS) dates and installation durations are assumed for teaching.

2 Basis and references

Document	Title	Where in the OTS
PFD-201	Process flow diagram	PFD tab
MEOH-PID-001 / 002 Rev A	EPC P&IDs (tags, lines, loop schedule, fail positions)	P&ID tab
MEOH-V1-PE-HMB-001	Heat and material balance	Process engineering tab
MEOH-V1-PE-LSL-001	Line sizing list	Process engineering tab
MEOH-V1-ME-EQL-001	Equipment list (design pressures and temperatures)	Mechanical tab
MEOH-V1-ME-DS-0nn	Equipment datasheets (one per item)	Mechanical tab
MEOH-V1-PE-RLB-001	Relief load basis	Relief & vent tab
MEOH-V1-IN-INL-001	Instrument list	Instrumentation & control tab
MEOH-V1-IN-CE-001	Cause and effect diagram	C&E matrix tab
MEOH-V1-IN-SRS-001	Safety requirements specification	SRS of SIF tab
MEOH-V1-EE-LL-001 / HAC-001	Electrical load list, hazardous-area classification	Electrical tab
MEOH-V1-SA-HZP-001	HAZOP report	HAZOP tab
MEOH-V1-PM-COM-001	Pre-commissioning and commissioning plan (companion)	this tab

Model design point (noise off): converter feed 1928 kNm³/h, make-up 365.6 kNm³/h, converter inlet 79.7 bar g, V-201 75 bar g / 40 °C, methanol 112.5 t/h, C-201 6.53 MW at 80 % speed, catalyst 208.3 t in four beds (25 / 41.6 / 58.3 / 83.3 t), PSV-201 set 88 bar g on V-201, PSHH-203 trip 85 bar g.

Design pressures, temperatures and datasheet numbers are taken from the equipment list MEOH-V1-ME-EQL-001 (Mechanical tab): Zone A — low-pressure side: V-201, C-201 suction, E-203 outlet piping — 88 bar g; Zone B — C-201 discharge to V-201 inlet: C-201 casing, E-202 both sides, E-201 both sides, E-204 tube side, R-201, E-203 tube side — 100 bar g.

3 Criticality and inspection codes

Code	Meaning
A	Long lead, on or near the critical path, or safety-critical (SIL-rated, high-pressure hydrogen boundary). Expedited; requisition first.
B	Engineered item with normal lead; requisition in the second wave.
C	Catalogue item, bought from the bulk material take-off.
H	Hold point: work stops until the purchaser’s inspector attends and releases it.
W	Witness point: the inspector is notified; work continues if nobody attends.
R	Review of records (certificates, reports, calculations).

4 Package register

Package	Title and tags	Scope	Key documents	Lead time (months, typical, assumed)	Crit.
PKG-01	**R-201 methanol converter with internals** Tags: R-201	Heavy-wall quench converter for hot hydrogen service, 4 adiabatic beds (catalyst 25 / 41.6 / 58.3 / 83.3 t), inlet distributor, bed support grids, 3 lozenge quench distributors above beds 2–4, outlet collector, catalyst dump nozzles, manways, nozzles for the 12 bed thermocouples TE-212 … TE-244 and the PDI-215 … 245 taps. Design 100 bar g / 335 °C, 2¼Cr-1Mo (MEOH-V1-ME-EQL-001).	• R-201 datasheet MEOH-V1-ME-DS-001 (see Mechanical tab) • MEOH-V1-ME-EQL-001 • MEOH-PID-001 • MEOH-V1-PE-HMB-001	14–20	A
PKG-02	**Catalyst, support media and loading** Tags: R-201 beds 1–4	Cu/ZnO/Al_2O_3 methanol synthesis catalyst: 208.3 t loaded (model bed masses 25 + 41.6 + 58.3 + 83.3 t; 184 m^3 nominal) plus a spare quantity (assumed 5 %), inert support balls, dense or sock loading service with dust removal and bed-level survey. Oxide (in-situ reduction) or pre-reduced (stabilised) form to be decided (query).	• R-201 datasheet MEOH-V1-ME-DS-001 (bed volumes, see Mechanical tab) • Model bed masses K.Wb • Transition operation tab §6 (reduction outline)	6–9	A
PKG-03	**C-201 loop circulator package** Tags: C-201, SC on FIC-201	Centrifugal circulator with MV motor and variable-speed drive (FIC-201 trims speed), dry gas seals and seal-gas panel, lube-oil console, coupling, baseplate, vibration and bearing-temperature monitoring. Design case (model): 1928 kNm^3/h, about 5.6 bar rise, 80 % speed, 6.53 MW shaft power; loop gas MW 5.83 kg/kmol. A nitrogen running case for dry-out and catalyst heat-up must be in the datasheet (query).	• C-201 datasheet MEOH-V1-ME-DS-007 (see Mechanical tab) • MEOH-V1-PE-HMB-001 (cases) • MEOH-V1-EE-LL-001 (motor, see Electrical tab) • MEOH-PID-002	12–18	A
PKG-04	**Feed/effluent interchangers E-201 and E-202** Tags: E-201, E-202	High-pressure gas/gas interchangers, both sides at loop pressure. Duties (model): E-201 44.2 MW, E-202 36.5 MW. Hot hydrogen on the E-201 hot side (effluent about 267–270 °C).	• E-201 datasheet MEOH-V1-ME-DS-003 • E-202 datasheet MEOH-V1-ME-DS-004 (see Mechanical tab) • MEOH-V1-PE-HMB-001 • MEOH-PID-001 / 002	10–14	A
PKG-05	**Final cooler E-203 and start-up heater E-204** Tags: E-203, E-204	E-203 cooling-water final cooler (CW 32 °C supply, separator 40 °C); E-204 HP-steam start-up heater (40 bar g steam, maximum 30 MW, used only for the catalyst heat-up).	• E-203 datasheet MEOH-V1-ME-DS-005 • E-204 datasheet MEOH-V1-ME-DS-006 (see Mechanical tab) • MEOH-V1-PE-RLB-001 (tube rupture, see Relief tab)	8–12	B
PKG-06	**V-201 high-pressure separator** Tags: V-201	Vertical separator with demister, 75 bar g / 40 °C, nozzles for PT-203, the PSHH-203 and LSHH-204 transmitters, LT-204, PSV-201 and HV-205. Design 88 bar g / 120 °C (= PSV-201 set, zone A, MEOH-V1-ME-EQL-001).	• V-201 datasheet MEOH-V1-ME-DS-002 (see Mechanical tab) • MEOH-PID-002 • MEOH-V1-PE-RLB-001	8–12	B
PKG-07	**Control valves** Tags: FV-202, PV-203, LV-204, TV-211, TV-221, TV-231, TV-241, HV-214	Globe control valves with pneumatic actuators and smart positioners; fail positions per the P&ID loop schedule; LV-204 severe let-down service (about 75 to 6 bar, flashing crude methanol, anti-cavitation trim); TV-211/221/231/241 fail open (converter goes colder on air failure).	• MEOH-V1-IN-INL-001 and the valve datasheets (see Instrumentation tab) • MEOH-V1-PE-LSL-001 (line sizes, see Process engineering tab) • MEOH-PID-001 / 002 loop schedule	5–10	B
PKG-08	**SIS logic solver SIS-201 and ESD console** Tags: SIS-201, HS-200	Safety PLC certified to IEC 61508 for the SIL 2 SIFs of the SRS, independent of the DCS, scan 0.25 s, de-energise to trip, latching trips with first-out and manual reset, start permissives PERM-1 … 6, maintenance bypass with alarm, ESD push-button HS-200. Application program from the C&E (39 rows).	• MEOH-V1-IN-SRS-001 (SRS of SIF tab) • MEOH-V1-IN-CE-001 (C&E matrix tab) • MEOH-V1-PM-COM-001 §3 (FAT test sheet)	6–9	A

Package	Title and tags	Scope	Key documents	Lead time (months, typical, assumed)	Crit.
PKG-09	**SIS field devices** Tags: XV-202, PSHH-203 transmitter, LSHH-204 transmitter, FT-201 (shared, HAZOP R01), C-201 stop relay	XV-202 make-up shut-off valve on DN300-SG-202 (fail closed, spring return, tight shut-off, solenoid de-energises to trip, stroke ≤ 2 s assumed); dedicated pressure and level transmitters for PSHH-203 and LSHH-204; options for FT-201B and dedicated bed elements (proposals in the SRS).	• MEOH-V1-IN-SRS-001 • MEOH-V1-IN-INL-001 (see Instrumentation tab) • MEOH-PID-002	6–10	A
PKG-10	**Relief and blowdown devices** Tags: PSV-201, HV-205 + RO, PSV-205, E-203 CW-side relief (TSV-203 + tube-rupture device), E-204 steam-side PSV (tag TBC)	PSV-201 (on the P&ID): V-201 top to flare, set 88 bar g, N orifice (4N6), governing case S2; HV-205 + RO (on the P&ID (RO size from RLB)): V-201 vapour to flare, manual blowdown, restriction orifice ≈ 49 mm; PSV-205 (proposed, HAZOP R03; not on the P&ID): let-down drum downstream of LV-204 (off-sheet), set 10 bar g (assumed), R (6R8) orifice; E-203 CW-side relief (TSV-203 + tube-rupture device) (proposed (RLB, Safety layers tab)): E-203 cooling-water shell, set 10 bar g (assumed), L (4L6) or bursting disc; TSV-203 DN20 × DN25 (thermal, typical); E-204 steam-side PSV (tag TBC) (proposed (RLB)): E-204 HP steam shell, set 45 bar g (assumed), H (2H3) orifice. Sizing and orifice letters per the relief load basis.	• MEOH-V1-PE-RLB-001 (see Relief & vent tab) • MEOH-PID-002 note 6 • E-203 datasheet MEOH-V1-ME-DS-005 • E-204 datasheet MEOH-V1-ME-DS-006	4–7	B
PKG-11	**Instruments and analyzers** Tags: FT/FE-201, 202, 203, 209, 212, 222, 232, 242; PT-201, 203; LT-204; TE-212 … 244 (multipoint); TI-202 … 254; PDT-215 … 245; AIT-205, AIT-208	Orifice flow elements and transmitters, pressure and level transmitters, multipoint bed thermocouple assemblies (12 points plus spares), temperature elements, differential-pressure transmitters across each bed, inlet CO/CO_2/CH_4 and outlet CH_3OH analyzers with sample systems and shelter; certified for the hazardous-area classification.	• MEOH-V1-IN-INL-001 (see Instrumentation tab) • MEOH-V1-EE-HAC-001 (see Electrical tab)	4–9	B
PKG-12	**DCS (basic process control system)** Tags: FIC-201, FIC-202, PIC-203, LIC-204, TIC-211, TIC-221, TIC-231, TIC-241, HIC-214; alarms; SIS interface	Controllers, operator stations and graphics for the 9 control loops, the alarm list, trends and the SIS-201 interface (status, first-out, bypass, reset); initial tuning from the simulator.	• MEOH-V1-IN-INL-001 • MEOH-PID-001 / 002 loop schedule • Alarm list (simulator ALM)	6–9	B
PKG-13	**Piping bulk materials** Tags: DN750-PG-201, DN300-PG-222, DN350-PG-232, DN350-PG-242, DN600-PG-213, DN900-PG-251, DN900-PG-205, DN150-PU-203, DN200-CM-209, DN300-SG-202, DN750-PG-202	Pipe, fittings, flanges (ring-type joints typical for high-pressure hydrogen service), gaskets, bolting and manual valves for the loop and its connections; large-bore heavy-wall forgings are the long-lead part. Quantities from the piping material take-off (not part of this teaching set).	• MEOH-V1-PE-LSL-001 (see Process engineering tab) • MEOH-PID-001 / 002	4–10	B
PKG-14	**Electrical equipment** Tags: C-201 drive supply, MV switchgear, transformer, LV MCC, cables, heat tracing, lighting, earthing	Power supply for the C-201 drive (the motor is in PKG-03), MV switchgear and transformer, LV distribution, cables, heat tracing, lighting and earthing; Ex equipment to the hazardous-area classification. Energisation is needed before C-201 can be run.	• MEOH-V1-EE-LL-001 (see Electrical tab) • MEOH-V1-EE-HAC-001	9–14	A

5 Inspection and test plan (summary)

Package	Title	Inspection and test activities (H / W / R)
PKG-01	R-201 methanol converter with internals	**R** · Material certificates EN 10204 3.2 for pressure parts in H_2 service; API 941 material selection **H** · Weld procedures and qualifications (WPS/PQR) approved before welding **W** · PMI of alloy parts and welds

Package	Title	Inspection and test activities (H / W / R)
		W · NDE: RT/UT of main seams and nozzle welds **R** · PWHT charts **H** · Shop hydrostatic test (ASME VIII-1 UG-99) **W** · Internals trial fit, dimensional check **R** · Manufacturer's data report and nameplate
PKG-02	Catalyst, support media and loading	**R** · Certificate of analysis per lot: CuO content, crush strength, size, loss on ignition **W** · Sampling at the works **H** · Loading of each bed: outage, density and level against the drawing **R** · Reduction (activation) procedure from the catalyst supplier
PKG-03	C-201 loop circulator package	**H** · Casing hydrostatic test, 1.5 × MAWP (API 617) **W** · Impeller overspeed test **W** · Mechanical running test at maximum continuous speed (API 617) **W** · Performance test (ASME PTC 10 type 2, typical) **W** · Lube-oil console and seal-gas panel function tests **R** · Rotor balance, vibration and noise records
PKG-04	Feed/effluent interchangers E-201 and E-202	**R** · Material certificates **W** · Tube-to-tubesheet weld mock-up and tests **W** · NDE **H** · Hydrostatic test of each side (ASME VIII-1) **W** · Bundle dimensional check **R** · Data report
PKG-05	Final cooler E-203 and start-up heater E-204	**R** · Material certificates **W** · NDE **H** · Hydrostatic test of each side **R** · Data report
PKG-06	V-201 high-pressure separator	**R** · Material certificates **W** · NDE **H** · Hydrostatic test **W** · Demister and internals inspection **R** · Data report
PKG-07	Control valves	**W** · Body hydrostatic shell test (ASME B16.34) **W** · Seat leakage test (ANSI/FCI 70-2 class per datasheet) **W** · Stroke, fail-action and positioner calibration test **R** · Sizing and noise calculations, material certificates
PKG-08	SIS logic solver SIS-201 and ESD console	**R** · Application program review against the SRS (IEC 61511-1 cl. 12) **H** · Integrated FAT: every C&E cause → effect, reset and permissive (MEOH-V1-PM-COM-001 §3) **W** · Hardware tests: redundant power, diagnostics, de-energise to trip **R** · Safety manual, certificates, FAT report and punch list
PKG-09	SIS field devices	**H** · XV-202 FAT: seat leakage, stroke time, fail action on loss of air and of power **R** · SIL certificates or FMEDA data of every device **W** · Transmitter calibration at the trip points
PKG-10	Relief and blowdown devices	**W** · Set-pressure and seat-tightness bench test of every PSV (API 527) **R** · Certified capacity and sizing check against the relief load basis **W** · HV-205 and restriction orifice: bore check, valve shell and seat test **R** · Material certificates

Package	Title	Inspection and test activities (H / W / R)
PKG-11	Instruments and analyzers	**W** · Calibration of transmitters **W** · Analyzer FAT with span gases **R** · Ex certificates and calibration records
PKG-12	DCS (basic process control system)	**H** · FAT: loop configuration, action, ranges, alarms and graphics against the P&ID **W** · Integrated test with SIS-201 **R** · FAT report and punch list
PKG-13	Piping bulk materials	**R** · Mill certificates **W** · PMI on alloy materials **W** · Manual valve pressure tests (API 598)
PKG-14	Electrical equipment	**W** · Routine tests of switchgear and transformer (IEC) **W** · Drive string test with the C-201 motor **R** · Test certificates

FAT procedures are submitted by each supplier for approval before the test; the SIS-201 FAT follows the test sheet in MEOH-V1-PM-COM-001 §3, generated from the C&E MEOH-V1-IN-CE-001.

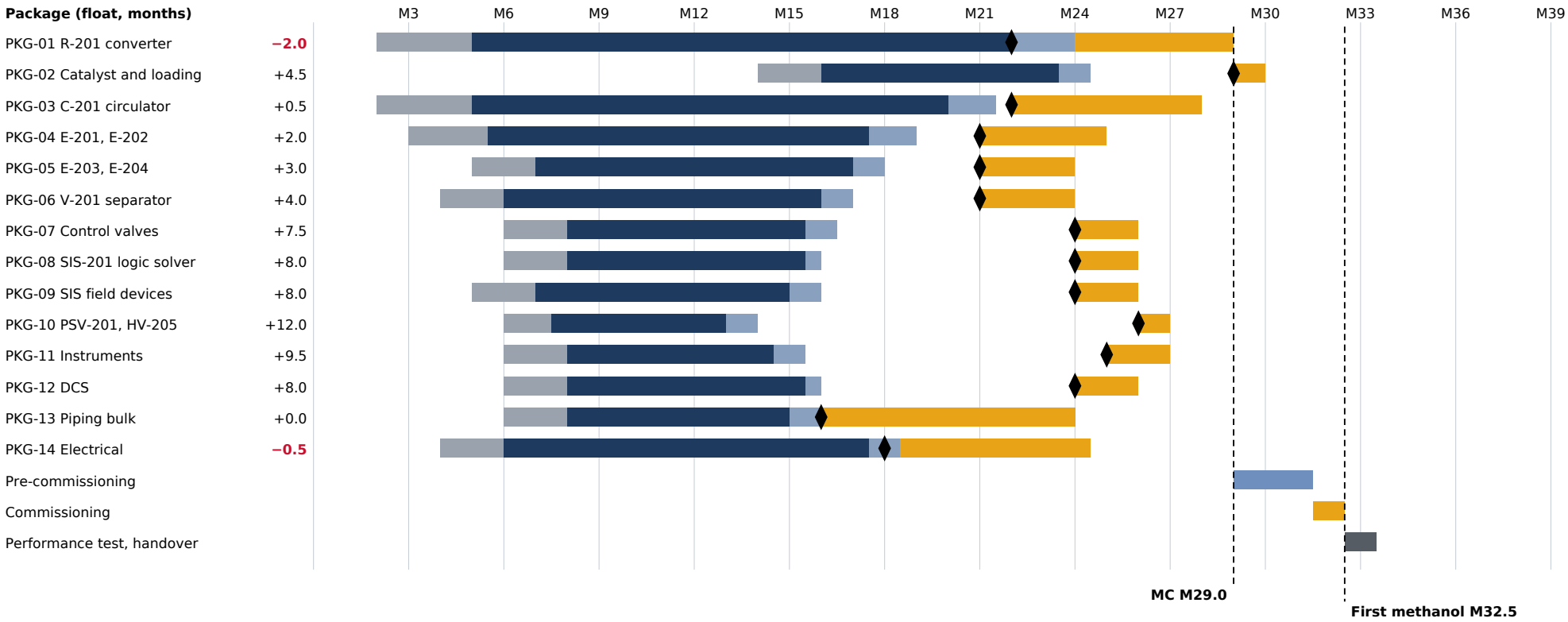
6 Schedule check against the required-on-site dates

Months from contract award (M0). Purchase order = requisition + bidding; ex-works = PO + manufacture (here the middle of the typical range); delivery = ex-works + shipping; installation starts at the later of delivery and ROS. The catalyst is needed at mechanical completion (MC) because it is loaded during pre-commissioning. MC = latest installation finish; ready for start-up = MC + 2.5 months of pre-commissioning (or catalyst delivery + 1.5 months if later); first methanol + 1.0 month; handover after a 1.0-month performance test. All durations assumed.

Package	Req.	PO	Manufacture (mo)	Ex-works	Delivery	ROS	Float (mo)	Install (mo)	Finish
PKG-01 R-201 methanol converter with internals	M2.0	M5.0	17.0	M22.0	M24.0	M22.0	−2.0	5	M29.0
PKG-02 Catalyst, support media and loading	M14.0	M16.0	7.5	M23.5	M24.5	M29.0 (= MC)	+4.5	1	M30.0
PKG-03 C-201 loop circulator package	M2.0	M5.0	15.0	M20.0	M21.5	M22.0	+0.5	6	M28.0
PKG-04 Feed/effluent interchangers E-201 and E-	M3.0	M5.5	12.0	M17.5	M19.0	M21.0	+2.0	4	M25.0
PKG-05 Final cooler E-203 and start-up heater E	M5.0	M7.0	10.0	M17.0	M18.0	M21.0	+3.0	3	M24.0
PKG-06 V-201 high-pressure separator	M4.0	M6.0	10.0	M16.0	M17.0	M21.0	+4.0	3	M24.0
PKG-07 Control valves	M6.0	M8.0	7.5	M15.5	M16.5	M24.0	+7.5	2	M26.0
PKG-08 SIS logic solver SIS-201 and ESD console	M6.0	M8.0	7.5	M15.5	M16.0	M24.0	+8.0	2	M26.0
PKG-09 SIS field devices	M5.0	M7.0	8.0	M15.0	M16.0	M24.0	+8.0	2	M26.0
PKG-10 Relief and blowdown devices	M6.0	M7.5	5.5	M13.0	M14.0	M26.0	+12.0	1	M27.0
PKG-11 Instruments and analyzers	M6.0	M8.0	6.5	M14.5	M15.5	M25.0	+9.5	2	M27.0
PKG-12 DCS (basic process control system)	M6.0	M8.0	7.5	M15.5	M16.0	M24.0	+8.0	2	M26.0
PKG-13 Piping bulk materials	M6.0	M8.0	7.0	M15.0	M16.0	M16.0	+0.0	8	M24.0
PKG-14 Electrical equipment	M4.0	M6.0	11.5	M17.5	M18.5	M18.0	−0.5	6	M24.5

Result (mid-range): MC M29.0, governed by PKG-01 R-201 methanol converter with internals; first methanol M32.5; handover M33.5. With every lead time at the short end of its range MC is M28.0; at the long end M32.0 (PKG-01). Packages arriving after their ROS date at mid-range: PKG-01 (−2.0 months), PKG-14 (−0.5 months).

Figure 1 · Package schedule at mid-range lead times



Bars: grey bidding · navy manufacture · light blue shipping · amber installation; ◆ required on site; float label red when the package arrives after its ROS date. The schedule explorer on the Procurement to commissioning tab moves the lead times, bidding time and late requisitions.

6a Relief and blowdown devices (PKG-10)

From the relief load basis MEOH-V1-PE-RLB-001. Proposed devices are ordered only once the RLB and the P&ID carry them.

Device	Where	Set	Size (RLB)	Status	Test before installation
PSV-201	V-201 top to flare	set 88 bar g	N orifice (4N6), governing case S2	on the P&ID	Bench set 88 bar g and seat tightness (API 527)
HV-205 + RO	V-201 vapour to flare	manual blowdown	restriction orifice ≈ 49 mm	on the P&ID (RO size from RLB)	Stroke and seat test; orifice bore checked against the RLB
PSV-205	let-down drum downstream of LV-204 (off-sheet)	set 10 bar g (assumed)	R (6R8) orifice	proposed, HAZOP R03; not on the P&ID	Bench set and seat tightness (API 527)
E-203 CW-side relief (TSV-203 + tube-rupture device)	E-203 cooling-water shell	set 10 bar g (assumed)	L (4L6) or bursting disc; TSV-203 DN20 × DN25 (thermal, typical)	proposed (RLB, Safety layers tab)	Bench set and seat tightness (API 527); bursting disc: certificate of burst pressure
E-204 steam-side PSV (tag TBC)	E-204 HP steam shell	set 45 bar g (assumed)	H (2H3) orifice	proposed (RLB)	Bench set and seat tightness (API 527)

7 Engineering queries

Raised while writing this list and the commissioning plan; they go to the engineering-query register of the Project & deliverables tab.

No.	From	To	Issue	Proposed	Status
EQ-ec-01	MEOH-V1-PM-COM-001	MEOH-V1-ME-EQL-001	Test pressures are computed from design pressures derived here by the typical rule (1.1 × maximum operating, and not less than PSV-201 set 88 bar g plus the loop pressure drop from the item to V-201): 94 bar g at C-201 discharge down to 88 bar g at V-201. The equipment list is not issued yet.	Confirm the design pressures and temperatures in the EQL and the datasheets; re-issue the pressure-test table. A single loop design pressure (the highest value) would simplify the piping class.	answered by MEOH-V1-ME-EQL-001 EQL zones: A (V-201, C-201 suction, E-203 outlet piping) 88 bar g = PSV-201 set; B (C-201 casing, E-202 both sides, E-201 both sides, E-204 tube side, R-201, E-203 tube side) 100 bar g (ASME UG-125: every item protected by PSV-201 ≤ 110 % of its MAWP → MAWP ≥ (1.1 × 88 + 11.3) / 1.1 = 98.2 bar g, rounded up to 5 bar). Test packs recomputed with these values and the EQL design temperatures.
EQ-ec-02	MEOH-V1-PM-PKG-001	R-201 datasheet / model header	Catalyst quantity: 184 m³ × 1150 kg/m³ = 211.6 t (model header), but the bed masses K.Wb add up to 208.3 t (also on MEOH-PID-001). The equipment list uses a loaded bulk density of 1131 kg/m³ (184.1 m³ → 208.3 t), consistent with K.Wb; the 1150 kg/m³ in the model header is not.	Order on volume (184 m³ plus a spare, assumed 5 %) with the loaded density confirmed by the catalyst supplier; correct one of the two numbers.	open

No.	From	To	Issue	Proposed	Status
EQ-ec-03	MEOH-V1-PM-PKG-001	C-201 datasheet (Mechanical tab)	C-201 is sized for loop gas of MW 5.83 kg/kmol. Dry-out and catalyst heat-up on nitrogen (MW 28) give about 4.8 × the gas density, so head and power at the same speed and pressure would overload the drive.	Add a nitrogen running case to the datasheet (reduced pressure about 1/4.8 of the loop pressure or reduced speed) and check the motor and drive before the PO.	open
EQ-ec-04	MEOH-V1-PM-PKG-001	MEOH-V1-IN-SRS-001	FSLL-201 shares FT-201 with FIC-201 and TSHH-2xx uses the indicating bed thermocouples (HAZOP R01, SRS proposals FSLL-201B and TSHH 2oo3). Extra elements change the R-201 nozzle schedule, which is frozen at the R-201 PO (critical path).	Decide on the SRS proposals before the R-201 requisition; include FT-201B and the dedicated thermocouple nozzles as priced options.	open
EQ-ec-05	MEOH-V1-PM-COM-001	MEOH-PID-002	The hand valves LB-204, PB-203 and BV-231 used in the SOPs and on the simulator's Field tab are not on the EPC P&IDs, so they are missing from the piping bulk and the stroke/line-up lists.	Add them to MEOH-PID-002 (LV-204 and PV-203 bypasses, BV-231) or remove them from the SOPs.	open
EQ-ec-06	MEOH-V1-PM-COM-001	MEOH-V1-PE-RLB-001	Hydrotest pressures (e.g. 114.4 bar g on V-201) exceed the PSV-201 set 88 bar g; the HV-205 restriction orifice has no size on the P&ID; E-203 CW side (10 bar g assumed) and E-204 steam side (45 bar g) are far below the loop pressure (tube-rupture case).	PSV-201 removed or blinded during the hydrotest and re-installed with its certificate; size the HV-205 orifice and settle the tube-rupture protection of E-203 and E-204 in the relief load basis before ordering PKG-05 and PKG-10.	open
EQ-ec-07	MEOH-V1-PM-COM-001	SOP tab (SOP-MEOH-01)	SOP-MEOH-01 assumes catalyst already reduced by the supplier procedure; the commissioning plan needs a decision between in-situ reduction (H_2 in N_2 circulation, low-pressure C-201 case, water drained at V-201) and pre-reduced catalyst.	Decide the catalyst form at the PKG-02 requisition; write the reduction SOP if in-situ.	open

8 Checks against the model and the other documents

Computed by epc/ecomm_data.js from the simulator model, the P&ID build scripts, the C&E data and the SOP register at issue; the tab recomputes them in the page.

Check	This document	Model or other document	Agrees
Valve fail positions: P&ID loop schedule vs model controller descriptions	FV-202 FC, PV-203 FC, LV-204 FC, TV-211 FO, TV-221 FO, TV-231 FO, TV-241 FO, HV-214 FC, HV-205 —, XV-202 FC	FV-202 FC, PV-203 FC, LV-204 FC, TV-211 FO, TV-221 FO, TV-231 FO, TV-241 FO, HV-214 FC, HV-205 —, XV-202 FC	✓
Every C&E trip and permissive is on the FAT sheet	HS-200, TSHH-2xx, FSLL-201, LSHH-204, PSHH-203, SIS-RST, PERM-1, PERM-2, PERM-3, PERM-4, PERM-5, PERM-6	HS-200, TSHH-2xx, FSLL-201, LSHH-204, PSHH-203, SIS-RST, PERM-1, PERM-2, PERM-3, PERM-4, PERM-5, PERM-6, ALARMS	✓
Every C&E alarm is on the DCS FAT	27 alarm rows	FAT-DCS-01	✓
Hydrotest pressure above design pressure (every item)	TP-01 150, TP-02 130, TP-03 130, TP-04 130, TP-05 130, TP-06 130, TP-07 130, TP-08 130, TP-09 114.4, TP-10 13, TP-11 58.5, TP-12 150, TP-15 132, TP-13 142.5, TP-14 15	design 100, 100, 100, 100, 100, 100, 100, 100, 88, 10, 45, 100, 88, 95, 10	✓
Loop design pressures ≥ PSV-201 set 88 bar g	100, 100, 100, 100, 100, 100, 100, 100, 88, 100, 88	PSV-201 88	✓
Design pressures taken from the equipment list MEOH-V1-ME-EQL-001	13 test packs	zone A 88, zone B 100 bar g	✓
EQL design pressure ≥ the typical rule (1.1 × op., PSV set + ΔP to V-201)	TP-01 100≥94, TP-02 100≥94, TP-03 100≥93, TP-04 100≥93, TP-05 100≥93, TP-06 100≥91, TP-07 100≥90, TP-08 100≥89, TP-09 88≥88, TP-12 100≥94, TP-15 88≥88	rule recomputed from the model pressures	✓
Leak-test pressure below 90 % of the PSV set	75 bar g	0.9 × 88 = 79.2	✓
PSHH-203 trip below PSV-201 set	85 bar g	88 bar g	✓
Every procedure referenced exists, step numbers within range	startup, opt, shutdown, hotstart, esd	6 OTS procedures	✓

Check	This document	Model or other document	Agrees
Every SOP referenced exists in the SOP register	SOP-MEOH-01, SOP-MEOH-02, SOP-MEOH-03, SOP-MEOH-04, SOP-MEOH-09, SOP-MEOH-10, SOP-MEOH-05, SOP-MEOH-07, SOP-MEOH-06, SOP-MEOH-12	12 SOPs	✓
Catalyst in PKG-02 = Σ model bed masses	208.3 t	Σ K.Wb = 208.3 t	✓
Catalyst volume × density (model header) = Σ K.Wb (EQ-ec-02)	184 m³ × 1150 kg/m³ = 211.6 t	Σ K.Wb = 208.3 t	✗
SIS-201 and DCS FAT run on copies of the model	23 model-tested rows	23 pass	✓

9 Notes

1. No vendor, client or contractor is named; package scopes are written for an open enquiry.

2. Lead times are typical ranges (assumed); they move with the market and must be confirmed in the bids. Heavy-wall hydrogen-service reactors, large compressors, MV drives and transformers are normally the long-lead items.

3. Spare parts (commissioning spares and two-year operating spares) and special tools are part of each package; capital spares (e.g. a spare C-201 rotor) are a separate decision.

4. Not for construction.